

DATASETS:

1. TOFU — primary dataset ★

[TOFU — Hugging Face download page](#)

This should be your **main dataset for the first prototype**. TOFU is specifically designed for LLM unlearning and contains QA pairs based on **200 fictional authors**. The authors and biographies are synthetic, so this is not a real-person privacy dataset •

Attributes / fields

TOFU is a **text QA dataset**. The important information in each example is essentially:

question

answer

The examples are questions and answers about the fictional authors.

For MindWipe:

- forget10 → INPUT TO UNLEARN
- retain90 → KNOWLEDGE TO PROTECT
- holdout → EVALUATION
- world_facts / real_authors → GENERALIZATION CHECK

2. MUSE — second-stage validation

[MUSE benchmark website](#)

MUSE is not just another simple forget/retain dataset. It is a **six-way evaluation benchmark** designed to test practical properties of machine unlearning. It evaluates:

MUSE attribute	What it tests
No verbatim memorization	Does the model reproduce removed text?
No knowledge memorization	Does it still retain the removed knowledge?
No privacy leakage	Can information still leak?
Utility preservation	Does normal model capability survive?
Scalability	Can larger deletion requests be handled?
Sustainability	Does repeated unlearning continue to work?

3. WMDP — robustness/safety evaluation

[WMDP — Hugging Face dataset](#)

WMDP stands for **Weapons of Mass Destruction Proxy**.

It contains multiple-choice questions designed to measure hazardous knowledge in:

WMDP

└─ Bio

└─ Cyber

└─ Chemistry

The public benchmark contains **3,668 questions** overall.

The Hugging Face dataset currently exposes configurations/splits including:

Split Purpose

bio Biosecurity knowledge

chem Chemical-security knowledge

cyber Cybersecurity knowledge

The dataset card reports **1,273 bio questions**, **408 chemistry questions**, and **1,987 cyber questions**.

WMDP is **multiple-choice**, rather than TOFU's open-ended QA format.

Conceptually:

- question
- choice A
- choice B
- choice C
- choice D
- correct answer
- category/domain

ALGORITHMS:

# Algorithm / technique	Why we use it	How it works — simplest explanation
1 SimNPO	Baseline	Pushes the model away from producing the forget-set answers , while avoiding some of the instability of conventional negative-gradient unlearning.
2 SIMU / Selective Neuron Updating	Selective baseline	Identifies neurons that matter for the forget data and updates only selected neurons , rather than modifying the whole model.
3 Mechanistic Localization + Causal Intervention	Find the <i>actual</i> components responsible for the target knowledge	Analyze activations to find candidate neurons/layers → temporarily intervene on them → check whether the forget behavior changes. Keep only candidates with useful causal effects.
4 PPO (Reinforcement Learning)	Adaptive decision maker	Given the validated candidate components, PPO chooses an intervention that maximizes forgetting while minimizing damage to retained knowledge.
5 Selective Parameter Update	Perform the final unlearning	Apply the chosen update only to the validated components instead of changing the entire LLM.